

PROBABILITY & STATISTICS FOR LIFE SCIENCES

Independence and Discrete Counting Models

Bernoulli, binomial, hypergeometric, and Poisson

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Purpose and standing assumptions

These notes continue from the properties of expectation and variance, develop independence through centered deviations, and build counting models from individual trials. Monday's material leads to the cereal-box example. The Wednesday continuation begins at [cereal part \(b\), page 16](#), derives binomial moments, contrasts sampling with and without replacement, and continues through hypergeometric and Poisson models to the end of Chapter 5.

All random variables in this article are **discrete and have finite second moments**. In particular, every expectation, variance, and covariance used below exists. Write $\mu_X = \mathbb{E}[X]$ and $\mu_Y = \mathbb{E}[Y]$. The notation $\text{Var}(X)$ means the same thing as $V(X)$ in the course slides.

The main thread is a question: *When may we add expectations, when may we add variances, and how do those facts help us understand a count of successes?* Keep the distinction between an individual outcome, a random count, and the probability of that count visible throughout.

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1 Independence: learning one value leaves the other distribution unchanged

Two random variables X and Y are independent when every pair of values has a joint probability that factors into its two marginal probabilities:

Definition of independence

$$\mathbb{P}(X = x, Y = y) = \mathbb{P}(X = x)\mathbb{P}(Y = y) \quad \text{for every pair } (x, y).$$

The word *every* matters. One factorization for one pair does not establish independence of the random variables.

Here $\mathbb{P}(X = x, Y = y)$ means the probability that *both* equalities hold. A marginal probability, such as $\mathbb{P}(Y = y)$, describes one variable without conditioning on the other. If $\mathbb{P}(X = x) > 0$, independence gives

$$\begin{aligned} \mathbb{P}(Y = y \mid X = x) &= \frac{\mathbb{P}(X = x, Y = y)}{\mathbb{P}(X = x)} && \text{definition of conditional probability} \\ &= \frac{\mathbb{P}(X = x)\mathbb{P}(Y = y)}{\mathbb{P}(X = x)} && \text{independence} \\ &= \mathbb{P}(Y = y). \end{aligned}$$

Thus, after learning $X = x$, the probabilities assigned to the possible values of Y stay the same. The corresponding statement also holds with X and Y reversed whenever the conditioning probability is positive.

Conversely, if the conditional distribution of Y is unchanged for every x with positive probability, multiplying by $\mathbb{P}(X = x)$ recovers the joint-probability definition. Values of x with probability zero cause no problem: all their joint probabilities are zero as well.

An information statement, not a causal statement

Independence describes a probability distribution. It says that learning one value does not change the distribution of the other. It does not, by itself, establish whether one physical process causes or influences another.

2 Linearity, products, and covariance

2.1 Expectation is linear without independence

For constants a, b, c , we always have

$$\mathbb{E}[aX + bY + c] = a\mathbb{E}[X] + b\mathbb{E}[Y] + c.$$

The two random variables may be independent, strongly related, or even the same random variable. None of those possibilities changes this rule. For example, $\mathbb{E}[X + X] = 2\mathbb{E}[X]$ is still linearity.

One way to see why dependence does not obstruct addition is to use the joint probability mass function.

Summing over all possible values,

$$\begin{aligned}
 \mathbb{E}[X + Y] &= \sum_x \sum_y (x + y) \mathbb{P}(X = x, Y = y) \\
 &= \sum_x x \underbrace{\sum_y \mathbb{P}(X = x, Y = y)}_{\mathbb{P}(X=x)} + \sum_y y \underbrace{\sum_x \mathbb{P}(X = x, Y = y)}_{\mathbb{P}(Y=y)} \\
 &= \mathbb{E}[X] + \mathbb{E}[Y].
 \end{aligned}$$

We only added the joint probabilities over the other variable's possible values. We did not factor the joint probability into a product.

2.2 Independence lets an expected product factor

For a product, a different step is needed. If X and Y are independent,

$$\begin{aligned}
 \mathbb{E}[XY] &= \sum_x \sum_y xy \mathbb{P}(X = x, Y = y) \\
 &= \sum_x \sum_y xy \mathbb{P}(X = x) \mathbb{P}(Y = y) && \text{independence is used here} \\
 &= \sum_x x \mathbb{P}(X = x) \left(\sum_y y \mathbb{P}(Y = y) \right) \\
 &= \left(\sum_x x \mathbb{P}(X = x) \right) \left(\sum_y y \mathbb{P}(Y = y) \right) \\
 &= \mathbb{E}[X] \mathbb{E}[Y].
 \end{aligned}$$

The inner sum no longer depends on x , so it can be taken outside the outer sum. Finite second moments ensure that these sums can be rearranged: in particular, $2|XY| \leq X^2 + Y^2$ gives $\mathbb{E}[|XY|] < \infty$.

Two rules with different requirements

$$\begin{aligned}
 \mathbb{E}[X + Y] &= \mathbb{E}[X] + \mathbb{E}[Y] && \text{always,} \\
 \mathbb{E}[XY] &= \mathbb{E}[X] \mathbb{E}[Y] && \text{if } X \text{ and } Y \text{ are independent.}
 \end{aligned}$$

Independence is sufficient for the product rule. We will soon see that it is not necessary.

2.3 Covariance records the expected product of deviations

The centered values $X - \mu_X$ and $Y - \mu_Y$ tell us how far each variable lies above or below its own mean. Their product is positive when their signs agree, negative when the signs differ, and zero when either deviation is zero. Covariance averages these products:

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)].$$

Expanding the expression gives a useful computational formula:

$$\begin{aligned}
 \text{Cov}(X, Y) &= \mathbb{E}[XY - \mu_Y X - \mu_X Y + \mu_X \mu_Y] \\
 &= \mathbb{E}[XY] - \mu_Y \mathbb{E}[X] - \mu_X \mathbb{E}[Y] + \mu_X \mu_Y \\
 &= \mathbb{E}[XY] - \mu_X \mu_Y - \mu_X \mu_Y + \mu_X \mu_Y \\
 &= \boxed{\mathbb{E}[XY] - \mathbb{E}[X] \mathbb{E}[Y]}.
 \end{aligned}$$

The means are fixed numbers, so linearity permits pulling them outside expectations. Independence was not used in this derivation.

3 Why variances add: inspect the cross term

3.1 Center first, then expand

Set

$$A = X - \mathbb{E}[X], \quad B = Y - \mathbb{E}[Y].$$

These centered deviations have mean zero:

$$\mathbb{E}[A] = \mathbb{E}[X] - \mathbb{E}[X] = 0, \quad \mathbb{E}[B] = 0.$$

The expectations of their squares recover the variances, $\mathbb{E}[A^2] = \text{Var}(X)$ and $\mathbb{E}[B^2] = \text{Var}(Y)$, while their expected product is $\mathbb{E}[AB] = \text{Cov}(X, Y)$.

Linearity gives $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$, so the deviation of the sum from its mean is exactly $A + B$. Now expand the square before taking averages:

$$\begin{aligned} \text{Var}(X + Y) &= \mathbb{E}[(X + Y) - \mathbb{E}[X + Y]]^2 \\ &= \mathbb{E}[(A + B)^2] \\ &= \mathbb{E}[A^2 + 2AB + B^2] \\ &= \mathbb{E}[A^2] + 2\mathbb{E}[AB] + \mathbb{E}[B^2] \\ &= \boxed{\text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)}. \end{aligned}$$

The extra term is not a technical afterthought: it measures whether the two deviations tend to reinforce or oppose one another.

For a difference, the centered deviation is $A - B$ instead:

$$\begin{aligned} \text{Var}(X - Y) &= \mathbb{E}[(A - B)^2] \\ &= \mathbb{E}[A^2 - 2AB + B^2] \\ &= \boxed{\text{Var}(X) + \text{Var}(Y) - 2\text{Cov}(X, Y)}. \end{aligned}$$

The individual squared terms remain positive; it is the *cross term* whose sign changes. We do not subtract $\text{Var}(Y)$.

3.2 The precise job of independence

If X and Y are independent, their centered versions A and B are also independent: subtracting a fixed number simply relabels the possible values. The product rule therefore gives

$$\mathbb{E}[AB] = \mathbb{E}[A]\mathbb{E}[B] = 0 \cdot 0 = 0.$$

This is why the cross term disappears. It does not require either distribution to be symmetric.

Independent sums and differences

When X and Y are independent,

$$\boxed{\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)}, \quad \boxed{\text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y)}.$$

The same formulas also hold whenever their covariance is zero, even if they are dependent.

3.3 Exactly what is equivalent to what?

For two variables with finite second moments, the identities we have derived show the exact equivalences

Equivalent statements for this pair of variables

$$\begin{aligned} & \text{Cov}(X, Y) = 0 \\ & \Updownarrow \\ & \mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y] \\ & \Updownarrow \\ & \text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) \\ & \Updownarrow \\ & \text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y). \end{aligned}$$

Independence implies all four statements. None of them, by itself, implies independence.

There is an important distinction between a *centered* product and a product of arbitrary random variables. For the centered A, B above, $\mathbb{E}[AB] = \text{Cov}(X, Y)$. To keep this separate from centering, use fresh names U, V for arbitrary random variables. In general,

$$\mathbb{E}[UV] = \text{Cov}(U, V) + \mathbb{E}[U]\mathbb{E}[V].$$

Thus $\mathbb{E}[UV] = 0$ holds exactly when the covariance and the product of the means cancel:

$$\text{Cov}(U, V) = -\mathbb{E}[U]\mathbb{E}[V].$$

Under independence, the covariance is zero, so $\mathbb{E}[UV] = 0$ holds exactly when at least one of $\mathbb{E}[U]$ and $\mathbb{E}[V]$ is zero. Without independence, even having *both* means zero is not enough.

4 Examples and the many-variable rule

4.1 Zero covariance can hide dependence

Let X be uniform on $\{-1, 0, 1\}$ and define $Y = X^2$. There are three possible joint outcomes, each with probability $1/3$:

X	$Y = X^2$	XY	Probability
-1	1	-1	1/3
0	0	0	1/3
1	1	1	1/3

Read each expectation as a weighted average of its column:

$$\begin{aligned}\mathbb{E}[X] &= (-1)\frac{1}{3} + 0\frac{1}{3} + 1\frac{1}{3} = 0, \\ \mathbb{E}[Y] &= 1\frac{1}{3} + 0\frac{1}{3} + 1\frac{1}{3} = \frac{2}{3}, \\ \mathbb{E}[XY] &= (-1)\frac{1}{3} + 0\frac{1}{3} + 1\frac{1}{3} = 0.\end{aligned}$$

Therefore

$$\text{Cov}(X, Y) = 0 - 0\left(\frac{2}{3}\right) = 0.$$

But knowing X determines Y completely. For example,

$$\mathbb{P}(Y = 0 \mid X = 0) = 1 \quad \text{whereas} \quad \mathbb{P}(Y = 0) = \frac{1}{3}.$$

The conditional distribution changes, so the variables are dependent. This one example also disproves the claim that $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$ implies independence. That equality checks one weighted average; independence checks the entire joint distribution.

4.2 Symmetry and zero means do not remove dependence

Let A take -1 and 1 with equal probabilities and let $B = A$. Both distributions are symmetric about zero, and $\mathbb{E}[A] = \mathbb{E}[B] = 0$. Nevertheless,

$$AB = A^2 = 1 \quad \text{on every outcome,} \quad \mathbb{E}[AB] = 1.$$

The deviations always have the same sign. Symmetry in each distribution does not make the two variables independent and does not force their expected product to vanish.

4.3 Several variables: each pair contributes a cross term

For $X_1, \dots, X_n$, expand the square of the sum of their centered deviations. Each squared term appears once; each distinct pair appears twice. Consequently,

$$\text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i) + 2 \sum_{1 \leq i < j \leq n} \text{Cov}(X_i, X_j).$$

The notation $i < j$ means that every pair is counted once. If the variables are independent, all pairwise covariances are zero and the variances add. In fact, *pairwise independence* is sufficient for this variance calculation, and pairwise zero covariance is sufficient as well. For three or more variables, variance additivity for the total only says that the *sum* of the cross covariances is zero; individual nonzero covariances can cancel.

4.4 Two independent copies versus twice the same variable

Suppose X_1 and X_2 are independent copies, each with variance σ^2 . Then

$$\text{Var}(X_1 + X_2) = \sigma^2 + \sigma^2 = 2\sigma^2.$$

If we double one variable instead, we double every deviation from its mean:

$$\text{Var}(2X) = \mathbb{E}[(2X - 2\mathbb{E}[X])^2] = 4\mathbb{E}[(X - \mathbb{E}[X])^2] = 4\sigma^2.$$

Equivalently, $\text{Cov}(X, X) = \text{Var}(X) = \sigma^2$, so adding X to itself includes the extra $2\sigma^2$ cross term.

Relationship	What we may conclude
Any X, Y	Expectations add; variances include $2 \text{Cov}(X, Y)$.
Independent X, Y	Expected products factor; covariance is zero; variances add.
$\text{Cov}(X, Y) = 0$	Expected products factor and variances add; independence is not established.
Symmetric distributions with zero means	No general conclusion that $\mathbb{E}[XY] = 0$.
Independent equal-variance copies	Variance of their sum is $2\sigma^2$.
Twice the same variable	Variance is $4\sigma^2$.

5 One trial: the Bernoulli distribution

5.1 Code the outcome as zero or one

A Bernoulli random variable records whether one specified event occurs. We call the event a “success” and code it as 1; otherwise the outcome is called a “failure” and coded as 0. These are neutral labels, not judgments about whether an outcome is desirable.

Write $p \in [0, 1]$ for the success probability and $q = 1 - p$ for the failure probability. Then $X \sim \text{Bernoulli}(p)$ has probability mass function

Outcome	Value x	$\mathbb{P}(X = x)$
Failure	0	$q = 1 - p$
Success	1	p

The expectation is the probability-weighted average of these numerical values:

$$\mathbb{E}[X] = 0 \cdot q + 1 \cdot p = p.$$

The mean equals p because of the zero–one coding.

5.2 Variance from squared deviations

Since the mean is p , the two deviations are $0 - p$ and $1 - p = q$. We square each deviation and weight it by the probability of its outcome:

Value	Deviation	Squared deviation	Weight	Contribution
0	$-p$	p^2	q	p^2q
1	$1 - p = q$	q^2	p	q^2p

Thus

$$\begin{aligned}
 \text{Var}(X) &= (0 - p)^2q + (1 - p)^2p \\
 &= p^2q + q^2p \\
 &= pq(p + q) && \text{factor out the common factor } pq \\
 &= \boxed{pq = p(1 - p)} && \text{because } p + q = 1.
 \end{aligned}$$

The squares and the probability weights have different jobs: p^2 is the squared deviation for outcome 0, while q is its probability.

5.3 A second derivation using the zero–one coding

For either possible outcome, $X^2 = X$: both $0^2 = 0$ and $1^2 = 1$. Consequently $\mathbb{E}[X^2] = \mathbb{E}[X] = p$. The variance shortcut gives

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = p - p^2 = p(1 - p).$$

Notice that $\mathbb{E}[X^2] = p$, whereas $(\mathbb{E}[X])^2 = p^2$. Taking a square before averaging and squaring an average are different operations.

5.4 When is a Bernoulli outcome most variable?

At $p = 0$, the outcome is certainly 0; at $p = 1$, it is certainly 1. In either case, the variance is zero because there is no uncertainty in the outcome. To find the largest variance, regard

$$V(p) = p(1 - p) = p - p^2, \quad 0 \leq p \leq 1,$$

as a function of p . Its derivatives are

$$V'(p) = 1 - 2p, \quad V''(p) = -2.$$

The derivative is zero at $p = 1/2$, and the negative second derivative shows that this is a maximum. Comparing with the zero-variance endpoints,

$$\boxed{\max_{0 \leq p \leq 1} p(1 - p) = \frac{1}{4}, \quad \text{attained at } p = \frac{1}{2}.}$$

A trial is most uncertain when its two outcomes are equally likely. The variance is $1/4$; the corresponding standard deviation is $1/2$.

6 Repeated trials: a binomial count

Fix a number of trials n . Each trial has two outcomes, the same success probability p , and the trials are *mutually independent*. Introduce an indicator for each trial:

$$X_i = \begin{cases} 1, & \text{trial } i \text{ is a success,} \\ 0, & \text{trial } i \text{ is a failure.} \end{cases} \quad X_i \sim \text{Bernoulli}(p).$$

The total number of successes is

$$\boxed{S = X_1 + \cdots + X_n \sim \text{Bin}(n, p).}$$

The symbol S here is a random *count*, taking values $0, 1, \dots, n$. It is not a probability. We use the upright letters S and F for the success and failure symbols in a sequence.

The model assumptions each matter: a fixed number of trials, independent outcomes, and a common success probability. Pairwise independence was enough to remove covariance terms, but the binomial model uses mutual independence to multiply probabilities along an entire outcome sequence.

6.1 Three coin flips: exactly two heads

Suppose three independent flips each have head probability p . There are three distinct sequences with exactly two heads:

Sequence	Probability
HHT	$p \cdot p \cdot q = p^2q$
HTH	$p \cdot q \cdot p = p^2q$
THH	$q \cdot p \cdot p = p^2q$

Independence lets us multiply within each sequence. The three sequences are disjoint outcomes, so we add their probabilities:

$$\mathbb{P}(\text{exactly two heads}) = p^2q + p^2q + p^2q = 3p^2q.$$

For a fair coin, this is $3(1/2)^2(1/2) = 3/8$. The factor 3 counts which positions contain the two heads.

7 Count the positions, then attach their probabilities

7.1 Five slots and exactly two successes: all ten arrangements

First ignore probabilities and ask only where the successes can go. Group the possible strings by the location of their first success:

First success in slot	Strings	Number
1	SSFFF, SFSFF, SFFSF, SFFFS	4
2	FSSFF, FSFSF, FSFFS	3
3	FFSSF, FFSFS	2
4	FFFSS	1
Total		$4 + 3 + 2 + 1 = 10$

Once the two success positions are chosen, every other slot must contain a failure. There is no further choice to make for those remaining slots.

7.2 Why dividing by a factorial removes overcounting

Temporarily distinguish the two successes by calling them S_A and S_B . There are five choices for the position of S_A and four remaining choices for the position of S_B :

$$5 \cdot 4 = 20 \quad \text{placements of the labeled successes.}$$

But the actual experiment records only success or failure, not an A or B label. For example,

$$S_A F S_B F F \quad \text{and} \quad S_B F S_A F F$$

both become the same string **SFSFF** when the labels are erased. Every success-position pattern was counted exactly $2! = 2$ times. Therefore the number of distinct success/failure strings is

$$\binom{5}{2} = \frac{5 \cdot 4}{2!} = 10.$$

The division does not discard possible outcomes. It corrects how often our temporary labeling counted each actual outcome.

For k labeled successes in n slots, the corresponding ordered count is

$$n(n-1) \cdots (n-k+1).$$

There are $k!$ ways to permute the labels among any fixed set of k success positions. Dividing by this common overcount gives

$$\begin{aligned} \binom{n}{k} &= \frac{n(n-1) \cdots (n-k+1)}{k!} \\ &= \boxed{\frac{n!}{k!(n-k)!}}. \end{aligned}$$

The factor $(n-k)!$ enters because $n(n-1) \cdots (n-k+1) = n!/(n-k)!$; it is not an additional choice of failure locations after the successes have been placed. At the endpoints, there is one all-failure string and one all-success string, so $\binom{n}{0} = \binom{n}{n} = 1$, consistent with $0! = 1$.

7.3 Match one probability factor to each letter

Return to **SFSFF**, this time with independent trials:

slot	1	2	3	4	5
outcome	S	F	S	F	F
probability factor	p	q	p	q	q

Thus the probability of this *particular ordered sequence* is

$$p \cdot q \cdot p \cdot q \cdot q = p^2 q^3.$$

Every one of the ten strings has two successes and three failures, so every one has that same probability. Adding the ten disjoint outcomes,

$$\mathbb{P}(S = 2) = 10p^2 q^3 = \binom{5}{2} p^2 q^3.$$

7.4 The general binomial probability mass function

The same reasoning works for any $k \in \{0, 1, \dots, n\}$:

1. A specified sequence with k successes and $n-k$ failures has probability $p^k q^{n-k}$, by independence.
2. There are $\binom{n}{k}$ such sequences.
3. They are disjoint and all have the same probability, so their probabilities add.

Therefore, for $0 < p < 1$,

Binomial probability mass function

$$\mathbb{P}(S = k) = \binom{n}{k} p^k q^{n-k}, \quad k = 0, 1, \dots, n, \quad q = 1 - p.$$

It is zero for other values of k . The coefficient counts the sequences; the powers give the probability of one sequence.

What is being multiplied?

The factor $p^k q^{n-k}$ comes from k individual success factors and $n - k$ individual *failure* factors in one specified sequence. It is not the product of two count-event probabilities such as “the probability of k successes” and “the probability of $n - k$ successes.” Exactly k successes already forces $n - k$ failures.

8 Why the binomial probabilities sum to one

8.1 A five-trial fair example

When $n = 5$ and $p = q = 1/2$, each of the $2^5 = 32$ success/failure strings has probability $1/32$. Grouping them by the number of successes gives

Success count k	0	1	2	3	4	5
Number of strings $\binom{5}{k}$	1	5	10	10	5	1
$\mathbb{P}(S = k)$	1/32	5/32	10/32	10/32	5/32	1/32

Every probability is nonnegative, and

$$\frac{1 + 5 + 10 + 10 + 5 + 1}{32} = \frac{32}{32} = 1.$$

These groups exhaust the possible strings and do not overlap.

8.2 Pascal's triangle counts the same groups

The rows below list $\binom{n}{k}$ as k runs from 0 to n :

Number of trials n	Counts in increasing order of k					
0	1					
1	1	1				
2	1	2	1			
3	1	3	3	1		
4	1	4	6	4	1	
5	1	5	10	10	5	1

To count length- n strings with k successes, look at their last slot. If the last outcome is a success, the first $n - 1$ slots must contain $k - 1$ successes. If it is a failure, the first $n - 1$ slots must contain k successes. The two cases are disjoint, giving, for $1 \leq k \leq n - 1$,

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

For instance, $\binom{5}{2} = \binom{4}{1} + \binom{4}{2} = 4 + 6 = 10$. The endpoint counts are both 1. The $n = 0$ row records the one empty sequence, which has zero successes.

8.3 The binomial theorem is the algebra of these choices

Expand a product of five identical factors:

$$(q + p)^5 = (q + p)(q + p)(q + p)(q + p)(q + p).$$

To obtain a term, choose either q or p from each parenthesis. A term with two choices of p and three choices of q has value p^2q^3 ; there are ten ways to make those choices. Combining equal terms gives

$$(q + p)^5 = q^5 + 5pq^4 + 10p^2q^3 + 10p^3q^2 + 5p^4q + p^5.$$

More generally, choosing the k parentheses that contribute p gives

$$(q + p)^n = \sum_{k=0}^n \binom{n}{k} p^k q^{n-k}.$$

Since $p + q = 1$, the sum of the binomial probabilities is

$$\sum_{k=0}^n \mathbb{P}(S = k) = \sum_{k=0}^n \binom{n}{k} p^k q^{n-k} = 1^n = 1.$$

For $0 < p < 1$, every term is nonnegative, so these numbers form a valid probability mass function.

8.4 Endpoint probabilities and the distinction from expectation

At $p = 0$, every trial fails: $\mathbb{P}(S = 0) = 1$, with all other count probabilities zero. At $p = 1$, every trial succeeds: $\mathbb{P}(S = n) = 1$, with all other probabilities zero. State these distributions directly rather than asking students to interpret 0^0 in the formula. If $n = 0$ is allowed, the count is zero with probability one for any p .

Normalization and expectation are different sums:

$$\underbrace{\sum_{k=0}^n \mathbb{P}(S = k) = 1}_{\text{add the probabilities}} \quad \text{versus} \quad \underbrace{\mathbb{E}[S] = \sum_{k=0}^n k \mathbb{P}(S = k)}_{\text{weight each value by its probability}}.$$

Proving that the probabilities sum to one has not yet computed the average number of successes. We will derive that average from indicators after examining the cereal example.

9 Read a binomial histogram and evaluate a decision rule

9.1 Each bar represents a possible count

For $X \sim \text{Bin}(n, p)$, the horizontal axis lists a count k , and the height of its bar is $\mathbb{P}(X = k)$. A group of bars represents an event: for example, $X \leq 4$ includes the bars at 0, 1, 2, 3, 4, and its probability is the sum of their heights. These are discrete masses; the gaps between the integer counts carry no probability.

For a fixed n , small p concentrates probability near small counts; large p concentrates it near large counts. The histogram need not be symmetric. A mean marker is a probability-weighted balance point, not necessarily a possible count and not a decision threshold.

9.2 The cereal company's claim and the chosen rule

Suppose a cereal company claims that *at least* 15% of its boxes contain prizes. Let p be the actual probability that a sampled box contains a prize. Treat the outcomes for the 50 sampled boxes as independent, each with the same probability p . Let X count the boxes with prizes, so $X \sim \text{Bin}(50, p)$.

The claim is $p \geq 0.15$, not necessarily $p = 0.15$. Consider the rule chosen before observing the sample:

Decision rule

$$X \leq 4 \implies \text{reject the claim,}$$

$$X \geq 5 \implies \text{do not reject the claim.}$$

“Do not reject” means this rule has not supplied enough evidence against the claim. It does not establish that the claim is true.

9.3 Cereal example, part (a): false rejection at the boundary of a true claim

First take the boundary value $p = 0.15$. Write $\mathbb{P}_{0.15}$ to indicate that probabilities are computed under this particular true prize rate. Then

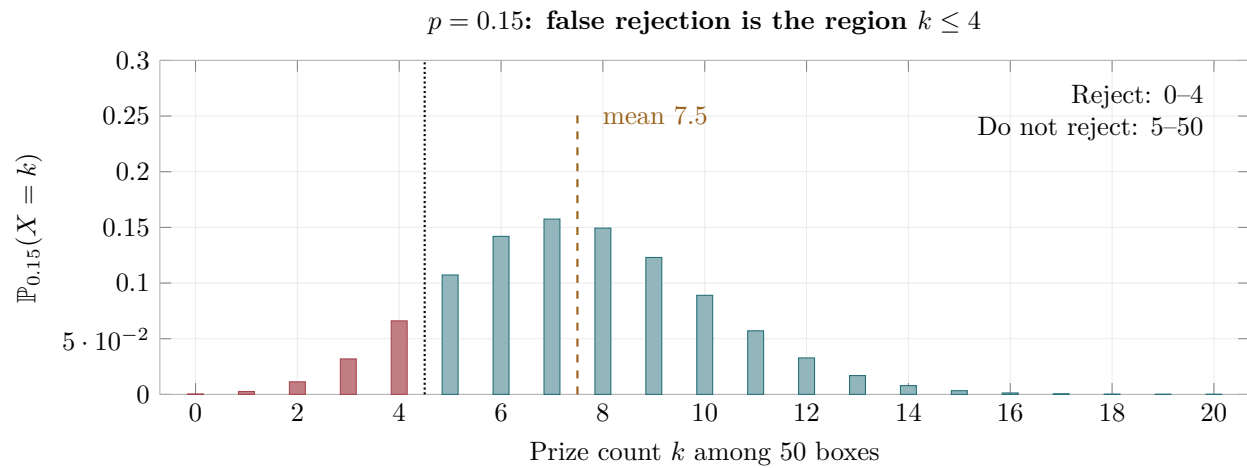
$$\begin{aligned} \mathbb{P}_{0.15}(\text{reject}) &= \mathbb{P}_{0.15}(X \leq 4) \\ &= \sum_{k=0}^4 \mathbb{P}_{0.15}(X = k) \\ &= \sum_{k=0}^4 \binom{50}{k} (0.15)^k (0.85)^{50-k} \\ &\approx \boxed{0.1121 \text{ or } 11.21\%}. \end{aligned}$$

At a true prize rate of 15%, repeated samples of 50 boxes would trigger this false rejection about 11.21% of the time.

Why use the boundary? As the true prize probability increases, low counts become less likely. One way to picture this is to turn some failure outcomes into successes while leaving all existing successes in place. The total count can only increase, so the event “at most four successes” cannot become more likely. Therefore

$$\mathbb{P}_p(X \leq 4) \leq \mathbb{P}_{0.15}(X \leq 4) \quad \text{for every } p \geq 0.15.$$

The value 0.1121 is the *maximum* false-rejection probability over all prize rates consistent with the claim. The actual probability may be smaller if the true rate exceeds 15%.



The red bars at 0–4 total 0.1121. The dotted divider sits between the two decisions; the dashed gold line marks the mean, derived below. For readability the plot shows 0–20 on the full support 0–50. The omitted probability $\mathbb{P}_{0.15}(X \geq 21)$ is about 3.90×10^{-6} and belongs to the do-not-reject region.

9.4 Cereal example, part (b): missing a false claim at a five-percent rate

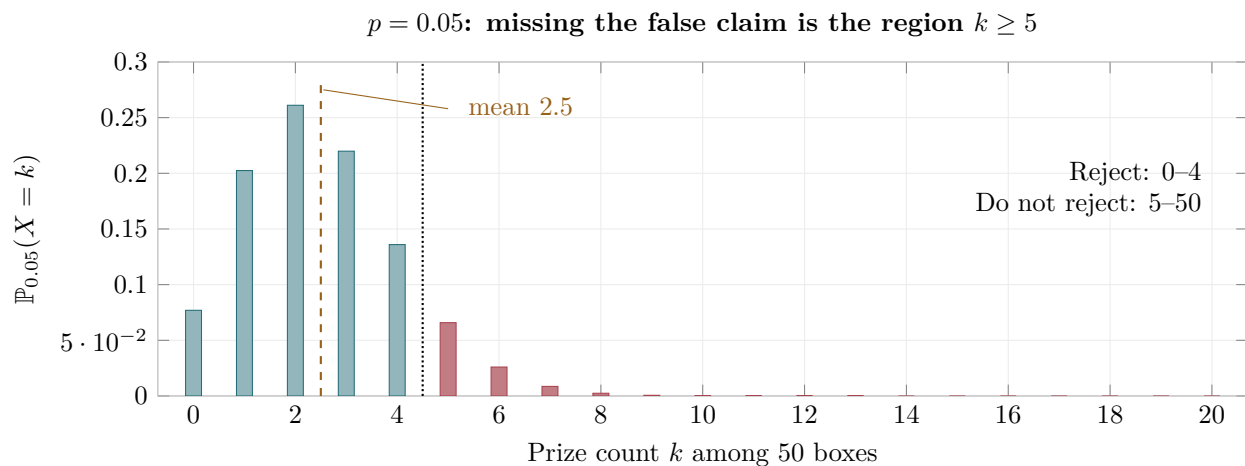
Wednesday starts here

Keep the same sample of 50 boxes and the same rule: reject when $X \leq 4$. We now change the assumed true prize rate to $p = 0.05$. After evaluating this second error, we will connect the prize count to a sum of Bernoulli variables and use that representation to derive its mean and variance.

Now change the assumed truth to $p = 0.05$. The sampling rule remains the same, but $X \sim \text{Bin}(50, 0.05)$. The company claim is false, and the error of interest is *not* rejecting it:

$$\begin{aligned}\mathbb{P}_{0.05}(\text{do not reject}) &= \mathbb{P}_{0.05}(X > 4) \\ &= 1 - \mathbb{P}_{0.05}(X \leq 4) \\ &= 1 - \sum_{k=0}^4 \binom{50}{k} (0.05)^k (0.95)^{50-k} \\ &\approx \boxed{0.1036 \text{ or } 10.36\%}.\end{aligned}$$

Because X is an integer, $X > 4$ means $X \geq 5$. At a true rate of 5%, this rule misses the false claim about 10.36% of the time.



The red region now begins at 5. Its total probability, including the unplotted counts 21–50, is 0.1036. The omitted probability is about 7.79×10^{-15} . Both histograms use the same horizontal and vertical scales so their shapes and decision regions can be compared directly.

The mean here is 2.5, but “above the mean” is not the same event as “do not reject.” Counts 3 and 4 are above 2.5 and still lead to rejection. The chosen cutoff, not the mean, determines the decision.

9.5 Two errors, computed under two different truths

Assumed true rate	Error made by the rule	Probability
$p = 0.15$	Reject a true claim: $X \leq 4$	0.1121
$p = 0.05$	Do not reject a false claim: $X \geq 5$	0.1036

These are different error types under different values of the true parameter. They are not complementary probabilities, and we do not add them to obtain one overall error probability without specifying how often the different true situations arise.

What these numbers do and do not answer

The first number is a probability of a decision *given a true prize rate*. It is not the probability that the company's claim is true or false after seeing a sample, nor the probability that a particular conclusion is wrong. Also, 11.21% alone does not establish that a sample of 50 boxes is insufficient. Adequacy depends on the sample size, the cutoff, the departures we want to detect, and which errors we are willing to tolerate.

For example, make rejection harder by changing the cutoff to $X \leq 3$. This removes the $X = 4$ bar from the rejection region. It reduces false rejection when the claim is true, but it also adds $X = 4$ to the region where a false claim is missed:

Rejection rule	False rejection at $p = 0.15$	Miss at $p = 0.05$
$X \leq 4$	0.1121	0.1036
$X \leq 3$	0.0460	0.2396

The sample size and the cutoff together determine the performance of the rule. This tradeoff is the intuitive bridge to later work on statistical testing; the binomial distribution already lets us quantify it.

10 A binomial count is a sum of Bernoulli variables

For the cereal experiment, define $I_i = 1$ when box i contains a prize and $I_i = 0$ otherwise. An outcome such as $(1, 0, 0, 1, 0)$ has two prize-bearing boxes because $1 + 0 + 0 + 1 + 0 = 2$. The equality

$$S = I_1 + \cdots + I_n$$

is true for every outcome. If the I_i are mutually independent and each is Bernoulli(p), any specified sequence with k ones has probability $p^k(1-p)^{n-k}$. There are $\binom{n}{k}$ such disjoint sequences, so

$$\mathbb{P}(S = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad S \sim \text{Bin}(n, p).$$

Independence identifies the distribution; the sum representation now gives us a simpler route to the moments than summing the binomial pmf directly.

10.1 Mean: add the expectations of the indicators

Return to the trial indicators. For $S \sim \text{Bin}(n, p)$,

$$S = X_1 + \cdots + X_n, \quad \mathbb{E}[X_i] = p, \quad \text{Var}(X_i) = pq.$$

Instead of directly evaluating the weighted sum $\sum_{k=0}^n k \binom{n}{k} p^k q^{n-k}$, use linearity:

$$\begin{aligned} \mathbb{E}[S] &= \mathbb{E}[X_1 + \cdots + X_n] \\ &= \mathbb{E}[X_1] + \cdots + \mathbb{E}[X_n] \\ &= \underbrace{p + \cdots + p}_{n \text{ terms}} \\ &= np. \end{aligned}$$

This addition of expectations does not require independence. Even if the trial outcomes were dependent, a sum of n indicators each having success probability p would still have mean np ; its distribution would not necessarily be binomial.

10.2 Variance: independence removes the cross covariances

For the binomial model the trial indicators are independent. Therefore

$$\begin{aligned}
 \text{Var}(S) &= \sum_{i=1}^n \text{Var}(X_i) + 2 \sum_{i < j} \text{Cov}(X_i, X_j) \\
 &= \sum_{i=1}^n pq + 0 && \text{independence makes the covariances zero} \\
 &= \boxed{npq = np(1-p)}.
 \end{aligned}$$

This is where independence matters in the moment calculation. Pairwise zero covariance would suffice for the variance identity, but independence is supplied by our binomial assumptions.

Binomial moments

$$S \sim \text{Bin}(n, p) \implies \mathbb{E}[S] = np, \quad \text{Var}(S) = np(1-p).$$

The mean adds because expectation is linear. The variances add because the independent indicators have zero cross covariances.

For the two cereal distributions,

Distribution	Mean prize count	Variance
Bin(50, 0.15)	$50(0.15) = 7.5$	$50(0.15)(0.85) = 6.375$
Bin(50, 0.05)	$50(0.05) = 2.5$	$50(0.05)(0.95) = 2.375$

A sample cannot contain 7.5 prize-bearing boxes. The fractional mean describes the long-run average count across repeated samples of 50 boxes under the model; it is neither a possible single-sample count nor a guarantee about the next sample. The variance describes the spread of those counts around their mean and is measured in squared count units.

Before changing the sampling model: which addition used independence?

In $\mathbb{E}[X_1 + \dots + X_n] = \sum_i \mathbb{E}[X_i]$, independence was not needed. In $\text{Var}(X_1 + \dots + X_n) = \sum_i \text{Var}(X_i)$, we used independence to remove the covariance terms. More precisely, zero cross covariances are sufficient. This is the distinction that turns the one-trial Bernoulli calculations into the binomial mean np and variance npq .

11 A fair coin connects the binomial formula to counting

We have obtained the binomial model by adding independent Bernoulli indicators. Its probability formula also connects directly to our earlier counting methods. Consider n independent flips of a fair coin, and let S count the heads. Here $p = 1/2$, so

$$\begin{aligned}
 \mathbb{P}(S = k) &= \binom{n}{k} \left(\frac{1}{2}\right)^k \left(\frac{1}{2}\right)^{n-k} \\
 &= \boxed{\frac{\binom{n}{k}}{2^n}}, \quad k = 0, 1, \dots, n.
 \end{aligned}$$

The two exponents add to n . Every complete sequence of n flips has probability $1/2^n$, and there are 2^n possible sequences. Exactly $\binom{n}{k}$ of them place heads in k positions.

11.1 Four flips: count the sequences with two heads

The six favorable sequences are

$$\text{HHTT, HTHT, HTTH, THHT, THTH, TTHH.}$$

There are $2^4 = 16$ equally likely sequences altogether. Therefore,

$$\mathbb{P}(S = 2) = \frac{6}{16} = \frac{3}{8}.$$

This is the same answer obtained from $\binom{4}{2}(1/2)^2(1/2)^2$. The binomial coefficient counts the favorable sequences; the factors of $1/2$ assign their probabilities.

When does favorable divided by total apply?

For independent fair flips, all 2^n complete sequences are equally likely, so counting favorable sequences and dividing by 2^n works. For an independent biased coin, a sequence with k heads still has probability $p^k(1-p)^{n-k}$, but sequences with different numbers of heads need not be equally likely. Keep the probability weights in that case.

Check. Why is $\mathbb{P}(S = k) = \binom{n}{k}/2^n$ a special case? It uses both independence of the flips and $p = 1/2$. Equal probabilities of heads and tails on each individual trial alone do not establish independence across trials.

12 Replacement connects the counting models

Now replace the coin with a finite collection of N distinct objects. Exactly K objects are labeled a success, and the other $N - K$ are labeled a failure. Draw n times. Let X be the number of successes drawn. The sampling procedure determines the distribution of X .

12.1 With replacement: the same composition on every draw

Suppose we select one of the N objects uniformly, record its label, replace it, and independently select again. Each draw has success probability

$$p = \frac{K}{N}.$$

Replacing the selected object restores the same collection. The independent uniform selections give independent Bernoulli trials, so

$$X \sim \text{Bin}\left(n, \frac{K}{N}\right).$$

There is a counting route to the same answer. With order recorded and replacement allowed, there are N^n equally likely object sequences. For a sequence with exactly k successes:

1. Choose the k success positions: $\binom{n}{k}$ choices.
2. Fill those positions using the K successful objects: K^k choices.
3. Fill the remaining positions using the $N - K$ unsuccessful objects: $(N - K)^{n-k}$ choices.

Thus

$$\begin{aligned}\mathbb{P}(X = k) &= \frac{\binom{n}{k} K^k (N - K)^{n-k}}{N^n} \\ &= \binom{n}{k} \left(\frac{K}{N}\right)^k \left(1 - \frac{K}{N}\right)^{n-k}.\end{aligned}$$

This is exactly the binomial formula. Replacement and independent selection allow the same object to appear more than once in the recorded sequence.

12.2 Without replacement: the remaining collection changes

If we keep each selected object out of the collection, an observed success changes what remains. For $0 < K < N$ and $N > 1$,

$$\mathbb{P}(\text{first draw is a success}) = \frac{K}{N},$$

whereas

$$\mathbb{P}(\text{second success} \mid \text{first success}) = \frac{K-1}{N-1} < \frac{K}{N}.$$

After a failure instead,

$$\mathbb{P}(\text{second success} \mid \text{first failure}) = \frac{K}{N-1} > \frac{K}{N}.$$

The label on the first draw therefore gives information about the next draw. The draw indicators are dependent, even though each draw position has the same success probability K/N before any outcomes are observed. We need a distribution that accounts for this sampling procedure.

13 The hypergeometric distribution: a count without replacement

Label the population and the sample separately

Symbol	Meaning
N	Total number of objects in the population
K	Number of successes in the population
n	Number of objects in the sample
k	A possible number of successes in that sample

The population has K successes and $N - K$ failures. A sample containing k successes contains $n - k$ failures. Here $N \geq 1$, $0 \leq K \leq N$, and $0 \leq n \leq N$, all integers.

Assume a *simple random sample without replacement*: every subset of n distinct population objects is equally likely. There are $\binom{N}{n}$ such samples.

To obtain exactly k successes, select k of the K successful objects and $n - k$ of the $N - K$ unsuccessful objects. Once those two selections are made, the sample is determined. Hence

$$\mathbb{P}(X = k) = \frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}.$$

We write

$$X \sim \text{Hypergeometric}(N, K, n).$$

The numerator counts the favorable samples. The denominator counts all equally likely samples. Both counts treat the sample as an unordered set of distinct objects.

13.1 Which values of the count are possible?

We cannot draw a negative number of successes or more than the K successes available. We also cannot draw more successes than the sample size n . Finally, if the population has only $N - K$ failures, a sample of size n must contain at least $n - (N - K)$ successes whenever that quantity is positive. Combining these restrictions gives

$$\max(0, n + K - N) \leq k \leq \min(n, K).$$

Outside this integer range the probability is zero.

Check. A population has $N = 10$ objects, of which $K = 8$ are successes. We draw $n = 5$ without replacement. The smallest possible success count is three, because only two failures exist. The largest is five. Thus the support is $\{3, 4, 5\}$.

13.2 Why the probabilities sum to one

Every sample has exactly one possible success count. Grouping all n -element samples by that count gives

$$\sum_k \binom{K}{k} \binom{N-K}{n-k} = \binom{N}{n},$$

where the sum runs over feasible k . Dividing by $\binom{N}{n}$ gives $\sum_k \mathbb{P}(X = k) = 1$. This is a counting proof of the normalization: the groups partition the set of all possible samples.

13.3 A balanced population is still different from independent fair flips

Take $N = 10$, $K = 5$, and $n = 4$. Each draw position has marginal success probability $1/2$. Without replacement,

$$\mathbb{P}(X = 2) = \frac{\binom{5}{2} \binom{5}{2}}{\binom{10}{4}} = \frac{100}{210} = \frac{10}{21} \approx 0.47619.$$

The independent fair-coin probability was $3/8 = 0.375$. Both experiments have success probability $1/2$ at each position, but their joint behavior differs. The without-replacement sample is more concentrated around its mean of two in this example.

13.4 Source example: how many purple candies?

A bag contains 50 candies of each of five colors, for 250 candies altogether. Select 12 uniformly without replacement. A success is selecting a purple candy. Therefore

$$N = 250, \quad K = 50, \quad n = 12.$$

Exactly half the sample being purple means $X = 6$, so

$$\mathbb{P}(X = 6) = \frac{\binom{50}{6} \binom{200}{6}}{\binom{250}{12}} \approx 0.0137602.$$

The second factor chooses six nonpurple candies, because a sample of twelve with six purple candies must have six of the other colors. We group all nonpurple colors together as failures.

14 Hypergeometric moments from the same indicator strategy

The count still has an indicator representation. Imagine the uniform sample being drawn sequentially without replacement, and define

$$I_i = \begin{cases} 1, & \text{draw } i \text{ is a success,} \\ 0, & \text{draw } i \text{ is a failure.} \end{cases} \quad X = I_1 + \cdots + I_n.$$

The indicators are dependent. We will retain the covariance terms instead of discarding them.

14.1 Mean: the marginal chance at each position is still K/N

Before any draws are observed, each of the N objects is equally likely to occupy a particular draw position. Exactly K objects are successes, so

$$\mathbb{P}(I_i = 1) = \frac{K}{N} =: p \quad \text{for every } i.$$

This is a marginal probability. It does not say the conditional chance stays fixed after earlier results are revealed.

By linearity,

$$\begin{aligned} \mathbb{E}[X] &= \mathbb{E}[I_1] + \cdots + \mathbb{E}[I_n] \\ &= \frac{K}{N} + \cdots + \frac{K}{N} \\ &= \boxed{n \frac{K}{N} = np}. \end{aligned}$$

The mean agrees with the binomial mean for the same n and p . The derivation remains valid because linearity never required independence.

14.2 Covariance: two successes compete for the finite supply

Assume $N > 1$. For two different draw positions i and j , the product $I_i I_j$ is one exactly when both positions contain successes. Uniform sampling without replacement gives

$$\mathbb{E}[I_i I_j] = \mathbb{P}(I_i = 1, I_j = 1) = \frac{K}{N} \frac{K-1}{N-1}.$$

When $K = 0$, the joint probability is zero directly; the displayed product has that same value. For $K > 0$, the second factor is the conditional chance after one success is used.

Use the covariance identity and put both terms over a common denominator:

$$\begin{aligned} \text{Cov}(I_i, I_j) &= \mathbb{E}[I_i I_j] - \mathbb{E}[I_i] \mathbb{E}[I_j] \\ &= \frac{K(K-1)}{N(N-1)} - \frac{K^2}{N^2} \\ &= \frac{NK(K-1) - K^2(N-1)}{N^2(N-1)} \\ &= -\frac{K(N-K)}{N^2(N-1)} \\ &= -\frac{p(1-p)}{N-1}. \end{aligned}$$

For $0 < K < N$, this covariance is negative. A success at one position leaves fewer successes available for another.

14.3 Variance: add the covariance contributions

Each I_i is marginally Bernoulli(p), so $\text{Var}(I_i) = p(1 - p)$. There are $\binom{n}{2}$ distinct unordered pairs of positions. Therefore

$$\begin{aligned}
 \text{Var}(X) &= \sum_{i=1}^n \text{Var}(I_i) + 2 \sum_{i < j} \text{Cov}(I_i, I_j) \\
 &= np(1 - p) + 2 \binom{n}{2} \left(-\frac{p(1 - p)}{N - 1} \right) \\
 &= np(1 - p) - \frac{n(n - 1)p(1 - p)}{N - 1} \\
 &= np(1 - p) \left(1 - \frac{n - 1}{N - 1} \right) \\
 &= \boxed{np(1 - p) \frac{N - n}{N - 1}}.
 \end{aligned}$$

Hypergeometric mean and variance

For $X \sim \text{Hypergeometric}(N, K, n)$, set $p = K/N$. Then

$$\mathbb{E}[X] = np, \quad \text{Var}(X) = np(1 - p) \underbrace{\frac{N - n}{N - 1}}_{\text{finite-population correction}} \quad (N > 1).$$

The mean uses linearity. The variance records the negative covariances created by sampling without replacement.

The correction provides useful checks:

- If $n = 1$, the correction is one. A single draw is Bernoulli(p).
- If $n = N$, the correction is zero. Inspecting the entire population gives exactly K successes, with no uncertainty in the count.
- If n is small relative to N , the correction is close to one, so the variance is close to the binomial variance.

If $N = 1$, handle the sample directly: the only possible sample sizes are zero and one, and its success count is fixed by the population. The formula with denominator $N - 1$ is not used in that case.

For the candy example,

$$\mathbb{E}[X] = 12 \left(\frac{50}{250} \right) = 2.4,$$

and

$$\text{Var}(X) = 12(0.2)(0.8) \frac{250 - 12}{250 - 1} = 1.92 \frac{238}{249} \approx 1.83518.$$

Independent sampling with replacement would have the same mean 2.4 and variance 1.92. Sampling without replacement reduces the variance.

15 Why hypergeometric counts approach binomial counts

Keep the sample size n fixed while the population becomes larger. If we remove only a few objects from a very large population, each removal makes a small change in the remaining success proportion. This suggests that sampling without replacement should resemble independent sampling with replacement.

Make the statement precise. Let N grow, allow $K = K_N$ to grow with it, and suppose

$$n \text{ is fixed, } \quad N \longrightarrow \infty, \quad \frac{K_N}{N} \longrightarrow p, \quad 0 < p < 1.$$

In particular, both the number of successes and the number of failures grow without bound. If $X_N \sim \text{Hypergeometric}(N, K_N, n)$, then

$$\mathbb{P}(X_N = k) \longrightarrow \binom{n}{k} p^k (1-p)^{n-k} \quad \text{for each } k = 0, 1, \dots, n.$$

Thus the distribution approaches $\text{Bin}(n, p)$.

15.1 Derive the limit directly from the counting formula

Write $a_{\underline{r}} = a(a-1)\cdots(a-r+1)$ for the falling factorial, with $a_0 = 1$. Expanding the binomial coefficients gives

$$\frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}} = \binom{n}{k} \frac{K_{\underline{k}} (N-K)_{\underline{n-k}}}{N_{\underline{n}}}.$$

For $K = K_N$, divide numerator and denominator by N^n :

$$\mathbb{P}(X_N = k) = \binom{n}{k} \frac{\prod_{j=0}^{k-1} \left(\frac{K_N}{N} - \frac{j}{N} \right) \prod_{j=0}^{n-k-1} \left(1 - \frac{K_N}{N} - \frac{j}{N} \right)}{\prod_{j=0}^{n-1} \left(1 - \frac{j}{N} \right)}.$$

An empty product equals one. Since n and k stay fixed, only finitely many factors appear. In the limit the first product becomes p^k , the second becomes $(1-p)^{n-k}$, and the denominator becomes one. That proves the displayed binomial limit.

The moments agree with this picture:

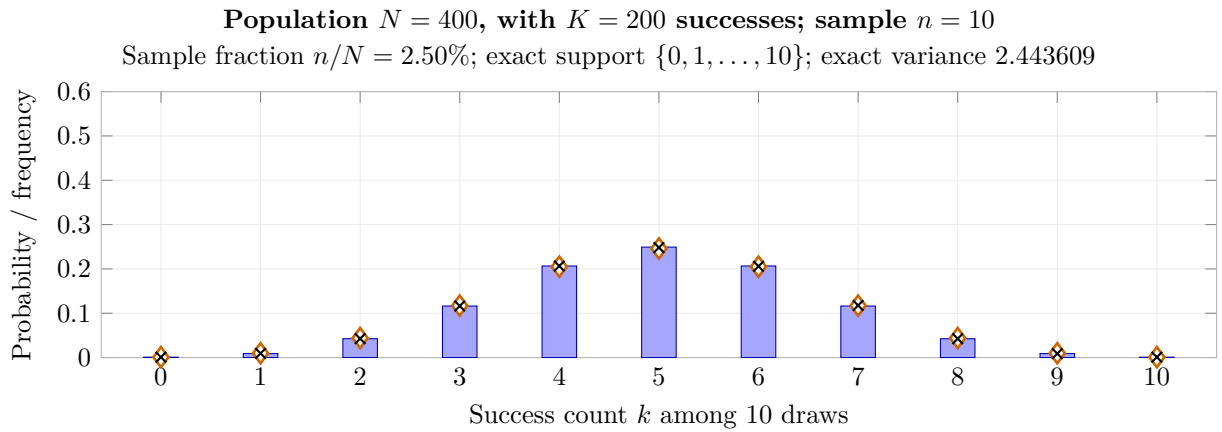
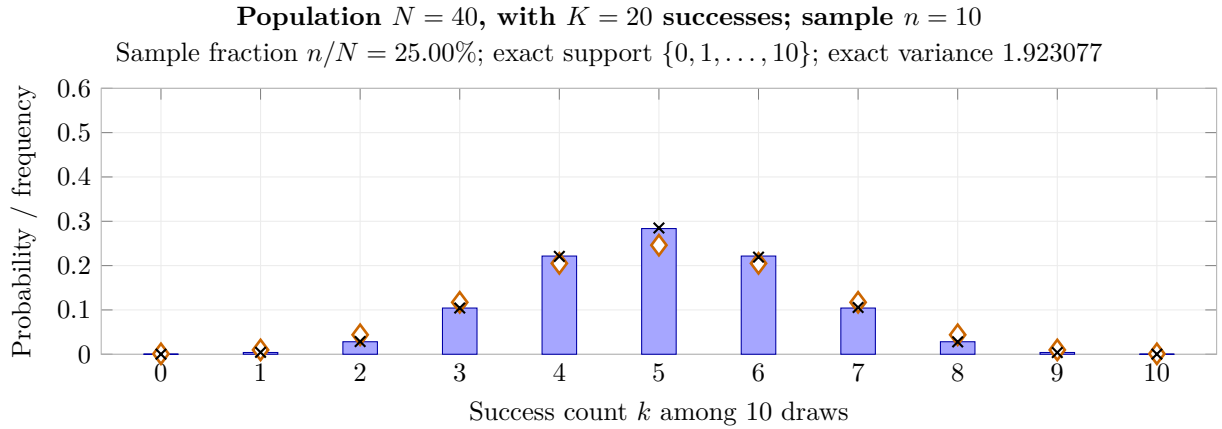
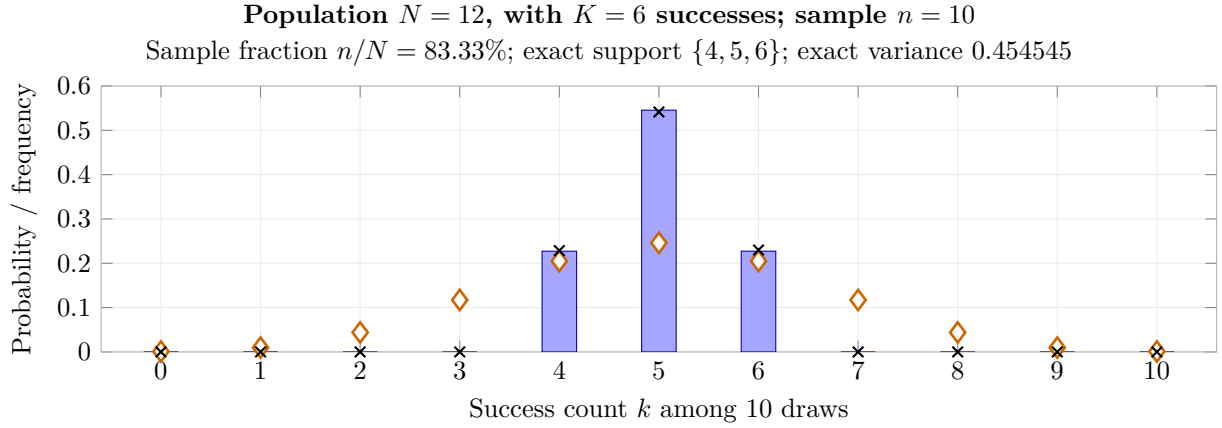
$$\begin{aligned} \mathbb{E}[X_N] &= n \frac{K_N}{N} \longrightarrow np, \\ \text{Var}(X_N) &= n \frac{K_N}{N} \left(1 - \frac{K_N}{N} \right) \frac{N-n}{N-1} \longrightarrow np(1-p). \end{aligned}$$

The correction tends to one because n is fixed while N increases.

A fixed sample from a growing population. Keep the sample size $n = 10$ and success fraction $p = K/N = 1/2$ fixed. Increase the population through $N = 12, 40, 400$, with $K = N/2$ successes. Sampling is without replacement within each sample; the population is reset before the next Monte Carlo repetition.

Blue bars are the **exact hypergeometric probabilities**. Orange open diamonds are the **exact $\text{Bin}(10, 1/2)$ probabilities**. Black crosses are **Monte Carlo relative frequencies** from 100,000

repetitions per panel. Marks refer only to integer counts, not a density. All panels have the same axes, show every count $k = 0, \dots, 10$, and omit no mass.



Measure the difference using exact probabilities. The common mean is $np = 5$. Sampling without replacement reduces the variance:

$$\text{Var}(X_N) = np(1-p) \frac{N-n}{N-1} = 2.5 \frac{N-10}{N-1} \rightarrow 2.5.$$

The total variation distance compares the two complete probability tables:

$$d_{\text{TV}}(X_N, B) = \frac{1}{2} \sum_{k=0}^{10} |\Pr(X_N = k) - \Pr(B = k)|, \quad B \sim \text{Bin}(10, 1/2).$$

It is the largest difference in probability the two models assign to the same event. Here it is calculated from exact formulas, not simulated counts.

N	K	Sample fraction n/N	Exact variance	Exact d_{TV}
12	6	83.33%	0.454545	0.343750
40	20	25.00%	1.923077	0.070407
400	200	2.50%	2.443609	0.006230
Binomial limit, $n = 10$, $p = 1/2$			2.500000	0

How to read the comparison. For $N = 12$, drawing 10 of the 12 objects forces the count into 4, 5, 6. The binomial model still assigns mass to $0, \dots, 10$, so it is a poor approximation. As N grows with n fixed, the sample fraction becomes small, the variance approaches 2.5, and the exact bars approach the orange marks. We are keeping n fixed: this is a binomial limit, not a normal approximation.

Simulation illustrates; it does not prove convergence. The three populations use seeds 20260942, 20260970, and 20261330 (base seed 20260930 plus N), with 100,000 repetitions each. The black crosses fluctuate around the exact blue bars. In the last panel, sampling noise can obscure the small remaining difference from the binomial model. The limit requires the mathematical PMF argument; neither a simulation nor three finite comparisons prove it.

15.2 What to look for in the comparison

Compare the probabilities assigned to each possible count k , keeping the sample size and limiting success proportion fixed. As the population grows, the hypergeometric probabilities should approach the binomial probabilities at every count. The smaller population has a more concentrated distribution because a larger fraction of it is sampled.

If a figure includes simulated frequencies, each repetition must start from the full original population, take a fresh sample without replacement, and record the success count. Across many repetitions, the relative frequencies estimate the hypergeometric pmf. Their random fluctuations are simulation noise; the exact pmf gives the underlying distribution to which those frequencies are compared.

Keep the sample size fixed in this limit

The statement above fixes n . Merely making N large does not make dependence negligible if the sample also occupies a substantial fraction of the population. The ratio $(N - n)/(N - 1)$ makes this distinction visible. For example, sampling half the population retains a correction near $1/2$ rather than near one.

16 Poisson begins with a concrete rare-event count

Suppose a laboratory processes 1000 specimens. Under a simplified model, each specimen independently experiences a particular rare processing error with probability 0.002. Let X count the specimens with that error.

The exact count model is

$$X \sim \text{Bin}(1000, 0.002).$$

Although there are many opportunities, the error probability at each opportunity is small. The expected number of errors is

$$\mathbb{E}[X] = 1000(0.002) = 2.$$

We can already answer concrete probability questions using the binomial formula:

$$\begin{aligned}\mathbb{P}(X = 0) &= (0.998)^{1000} \approx 0.135065, \\ \mathbb{P}(X \geq 1) &= 1 - (0.998)^{1000} \approx 0.864935, \\ \mathbb{P}(X = 2) &= \binom{1000}{2} (0.002)^2 (0.998)^{998} \approx 0.270942.\end{aligned}$$

The variance is $1000(0.002)(0.998) = 1.996$, which is close to the mean of two.

Now imagine more opportunities, each with a smaller error probability, while keeping the expected total at two. For example, 2000 independent opportunities with probability 0.001 again have expected count two. This is the setting behind the Poisson approximation: many small, independent chances combine into a count with a moderate expected value.

The parameter records the expected total

For a rare-event binomial approximation, set

$$\lambda = np.$$

The laboratory example has $\lambda = 2$. The parameter describes the expected number of events in the entire specified collection of opportunities.

17 The Poisson probability mass function

A Poisson random variable with parameter $\lambda > 0$ has pmf

$$\mathbb{P}(Y = k) = e^{-\lambda} \frac{\lambda^k}{k!}, \quad k = 0, 1, 2, \dots$$

We write $Y \sim \text{Poisson}(\lambda)$. This distribution gives probabilities to nonnegative integer counts. Its support has no finite upper bound.

For the laboratory example, use $Y \sim \text{Poisson}(2)$. The resulting approximations are

$$\begin{aligned}\mathbb{P}(X = 0) &\approx \mathbb{P}(Y = 0) = e^{-2} \approx 0.135335, \\ \mathbb{P}(X \geq 1) &\approx 1 - e^{-2} \approx 0.864665, \\ \mathbb{P}(X = 2) &\approx e^{-2} \frac{2^2}{2!} \approx 0.270671.\end{aligned}$$

Compare these with the exact binomial values above. The approximation is close for these probabilities. It replaces the separate parameters n and p with their product $\lambda = np$.

Approximation and exact model are different statements

The laboratory model is exactly binomial under the assumptions stated. Poisson provides an approximation here. A Poisson variable can exceed 1000, whereas the binomial count cannot; in this example the Poisson probability of such counts is extremely small. Keep the exact model and the approximation distinct.

17.1 Why do the Poisson probabilities sum to one?

Recall the exponential series:

$$e^\lambda = \sum_{k=0}^{\infty} \frac{\lambda^k}{k!}.$$

All the Poisson probabilities are nonnegative, and

$$\begin{aligned} \sum_{k=0}^{\infty} \mathbb{P}(Y = k) &= e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} \\ &= e^{-\lambda} e^\lambda \\ &= 1. \end{aligned}$$

Thus the formula defines a valid pmf. Also $0! = 1$, so the zero-count probability is simply $\mathbb{P}(Y = 0) = e^{-\lambda}$. The case $\lambda = 0$ can be included separately as the deterministic count $Y = 0$.

18 Derive Poisson as a limit of binomial distributions

Fix $\lambda > 0$. For each sufficiently large integer m , consider

$$Y_m \sim \text{Bin}\left(m, \frac{\lambda}{m}\right).$$

The number of trials is m , the success chance is λ/m , and the expected count is always λ . We use m here to emphasize that this number of opportunities is changing.

For a fixed count k and sufficiently large m ,

$$\mathbb{P}(Y_m = k) = \binom{m}{k} \left(\frac{\lambda}{m}\right)^k \left(1 - \frac{\lambda}{m}\right)^{m-k}.$$

Rearrange the factors:

$$\begin{aligned} \mathbb{P}(Y_m = k) &= \frac{m(m-1)\cdots(m-k+1)}{m^k} \frac{\lambda^k}{k!} \\ &\quad \cdot \left(1 - \frac{\lambda}{m}\right)^m \left(1 - \frac{\lambda}{m}\right)^{-k}. \end{aligned}$$

Each of the k ratios in the first factor approaches one. The standard exponential limit gives

$$\left(1 - \frac{\lambda}{m}\right)^m \longrightarrow e^{-\lambda},$$

and the final factor approaches one because k is fixed. Hence

$$\boxed{\mathbb{P}(Y_m = k) \longrightarrow e^{-\lambda} \frac{\lambda^k}{k!}.$$

For $k = 0$, the empty product in the first factor is one.

More generally, if $m \rightarrow \infty$ and $mp_m \rightarrow \lambda > 0$, then $p_m \rightarrow 0$ and $\text{Bin}(m, p_m)$ approaches $\text{Poisson}(\lambda)$. One can see the same limit by replacing λ in the calculation above with mp_m : $(mp_m)^k \rightarrow \lambda^k$ and $m \log(1 - p_m) \rightarrow -\lambda$.

18.1 Compare a source example numerically

For the source's parameters $m = 100$ and $p = 0.05$, use $\lambda = 5$.

k	Exact Bin(100, 0.05) probability	Poisson(5) probability
0	0.00592	0.00674
1	0.03116	0.03369
2	0.08118	0.08422
3	0.13958	0.14037
4	0.17814	0.17547

The expected count is five in both models. The quality of the approximation depends on the parameters and the probability being estimated. There is no need to approximate when an exact binomial calculation is convenient.

19 Poisson counts over time, area, or volume

The rare-opportunity picture also motivates a model for events over a fixed exposure. Suppose a detector records events at a constant rate r per minute. For an observation lasting t minutes, define

$$\lambda = rt.$$

The rate r has units of events per minute. The parameter λ is the expected number of events in this particular window.

19.1 State the assumptions before assigning the model

A homogeneous Poisson model for event times uses these properties:

- Counts in disjoint time intervals are independent.
- The rate r stays constant throughout the observation.
- Over a very short interval of length h , the probability of one event is approximately rh , while the probability of two or more events is negligible compared with h .

Under these assumptions, the count in a window of length t is $\text{Poisson}(rt)$.

To connect this to the binomial construction, divide the *same* window into m short pieces. Each piece has duration t/m and approximately $rt/m = \lambda/m$ chance of an event. The number of pieces containing an event is approximately $\text{Bin}(m, \lambda/m)$. Refining the pieces makes multiple events in a single piece negligible, so the occupied-piece count approaches the event count. The observation window stays fixed throughout.

Refining the pieces does not extend the observation

More pieces means more opportunities with a smaller chance per opportunity. The total expected count remains $m(\lambda/m) = \lambda$. Increasing m with a fixed positive success probability instead would make the expected count grow without bound.

19.2 A worked example with changing exposure

Suppose the detector model is appropriate with rate $r = 3$ events per minute. For the count Z in one minute, $\lambda = 3$:

$$\mathbb{P}(Z = 2) = e^{-3} \frac{3^2}{2!} \approx 0.224042,$$

$$\mathbb{P}(Z \geq 1) = 1 - \mathbb{P}(Z = 0) = 1 - e^{-3} \approx 0.950213.$$

For a 30-second observation, first convert the duration to $t = 0.5$ minute. The count W then has parameter $\lambda = 3(0.5) = 1.5$, so

$$\mathbb{P}(W \geq 1) = 1 - e^{-1.5} \approx 0.776870.$$

For two minutes the parameter is six. Changing the window changes the count distribution even though the event rate remains three per minute.

19.3 Read the source's examples as models with assumptions

The source uses an amoeba reproduction illustration. Its mathematical point is an idealized source generating events over a fixed interval $[t_0, t]$ with a constant rate and independent increments. Under the small-interval condition above, the count has parameter $r(t - t_0)$. This is an event-source model, not a literal claim that one amoeba repeatedly survives binary fission. Counting events from a growing collection of descendants would require reassessing the rate and independence assumptions.

The same exposure idea applies to the other source illustrations:

- **Raindrops:** count drops striking a specified area during a specified interval. With a homogeneous rate per unit area per unit time, multiply that rate by the area and duration.
- **Price jumps:** define what qualifies as one jump and count such events during a specified week. A constant-rate Poisson model also requires independent counts over disjoint intervals; a shared trigger or clustering can make that assumption unsuitable.
- **Stars:** count objects inside a fixed volume. With homogeneous spatial intensity ρ , the parameter is $\lambda = \rho V$. For the source's 500 cubic light-years, it is 500ρ . Independent counts in disjoint regions are part of the model.

These settings suggest potential applications. They do not by themselves establish the constant-intensity and independence assumptions.

20 Poisson expectation and variance

The binomial limit suggests the answer. For $Y_m \sim \text{Bin}(m, \lambda/m)$,

$$\mathbb{E}[Y_m] = \lambda, \quad \text{Var}(Y_m) = \lambda \left(1 - \frac{\lambda}{m}\right) \rightarrow \lambda.$$

This motivates the Poisson moment formulas. We now verify them directly from the pmf.

20.1 Expectation: cancel one factor and reindex

Let $Y \sim \text{Poisson}(\lambda)$. Then

$$\begin{aligned}
 \mathbb{E}[Y] &= \sum_{k=0}^{\infty} k e^{-\lambda} \frac{\lambda^k}{k!} \\
 &= e^{-\lambda} \sum_{k=1}^{\infty} \frac{\lambda^k}{(k-1)!} \\
 &= \lambda e^{-\lambda} \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!} \\
 &= \lambda e^{-\lambda} \sum_{j=0}^{\infty} \frac{\lambda^j}{j!} \\
 &= \lambda e^{-\lambda} e^{\lambda} \\
 &= \boxed{\lambda}.
 \end{aligned}$$

The $k = 0$ term contributes zero. For $k \geq 1$, $k/k! = 1/(k-1)!$. Taking out one factor of λ and setting $j = k - 1$ reveals the exponential series again.

20.2 Variance: use a product that cancels two factorial factors

The product $k(k-1)$ cancels neatly with $k!$, so first calculate $\mathbb{E}[Y(Y-1)]$:

$$\begin{aligned}
 \mathbb{E}[Y(Y-1)] &= \sum_{k=2}^{\infty} k(k-1) e^{-\lambda} \frac{\lambda^k}{k!} \\
 &= \lambda^2 e^{-\lambda} \sum_{k=2}^{\infty} \frac{\lambda^{k-2}}{(k-2)!} \\
 &= \lambda^2 e^{-\lambda} \sum_{j=0}^{\infty} \frac{\lambda^j}{j!} \\
 &= \lambda^2.
 \end{aligned}$$

The omitted terms $k = 0$ and $k = 1$ are both zero. Now write $Y^2 = Y(Y-1) + Y$. Linearity gives

$$\mathbb{E}[Y^2] = \mathbb{E}[Y(Y-1)] + \mathbb{E}[Y] = \lambda^2 + \lambda.$$

Finally,

$$\begin{aligned}
 \text{Var}(Y) &= \mathbb{E}[Y^2] - (\mathbb{E}[Y])^2 \\
 &= (\lambda^2 + \lambda) - \lambda^2 \\
 &= \boxed{\lambda}.
 \end{aligned}$$

Poisson moments

$$Y \sim \text{Poisson}(\lambda) \implies \mathbb{E}[Y] = \lambda, \quad \text{Var}(Y) = \lambda, \quad \text{SD}(Y) = \sqrt{\lambda}.$$

Mean and variance have the same numerical value under this model, but their units differ: counts for the mean and squared counts for the variance.

The detector count over two minutes has mean and variance six. Doubling exposure doubles both quantities under the homogeneous Poisson model. The standard deviation increases by a factor of $\sqrt{2}$, since it is the square root of the variance.

Check. If a Poisson count has mean four, its variance is four and its standard deviation is two. A Poisson assumption is stronger than merely asserting this relation between moments; it specifies every probability in the pmf.

21 Final Chapter 5 example: infer the parameter from a probability ratio

Suppose an annual insurance-claim count $X \sim \text{Poisson}(\lambda)$, with $\lambda > 0$, satisfies the following information: two claims are three times as likely as four claims. Find the probability of three or more claims next year under the same model.

Translate the statement before substituting the formula:

$$\mathbb{P}(X = 2) = 3\mathbb{P}(X = 4).$$

Thus

$$e^{-\lambda} \frac{\lambda^2}{2!} = 3e^{-\lambda} \frac{\lambda^4}{4!}.$$

Since $\lambda > 0$, cancel $e^{-\lambda}\lambda^2$:

$$\frac{1}{2} = \frac{3\lambda^2}{24} \implies \lambda^2 = 4 \implies \lambda = 2.$$

Only the positive root is an admissible parameter here. Now use the complement of the requested tail:

$$\begin{aligned} \mathbb{P}(X \geq 3) &= 1 - \mathbb{P}(X \leq 2) \\ &= 1 - (\mathbb{P}(X = 0) + \mathbb{P}(X = 1) + \mathbb{P}(X = 2)) \\ &= 1 - e^{-2} \left(1 + 2 + \frac{2^2}{2!} \right) \\ &= \boxed{1 - 5e^{-2} \approx 0.323324}. \end{aligned}$$

The first step identifies the parameter from a relation between probabilities. The second uses that parameter to calculate the requested probability. This is the final example in the canonical Chapter 5 notes.

22 Choose the experiment, then choose the count model

Experiment and count	Mean	Variance
$\text{Bin}(n, p)$: fixed n , independent binary trials, common p	np	$np(1 - p)$
$\text{Hypergeometric}(N, K, n)$: uniform sample without replacement; $p = K/N$	np	$np(1 - p) \frac{N - n}{N - 1}$ for $N > 1$
$\text{Poisson}(\lambda)$: event count under a Poisson model for a specified exposure	λ	λ

All three distributions describe counts. Their assumptions concern how the counts are produced. The indicator method explains their connections: expectations add without independence, while variance also reflects how the indicators vary together.

In the hypergeometric-to-binomial limit, keep the sample size fixed and enlarge the population. In the binomial-to-Poisson limit, enlarge the number of opportunities, shrink the success probability, and keep the expected count finite. These are different limiting experiments, so keep track of what is fixed in each.